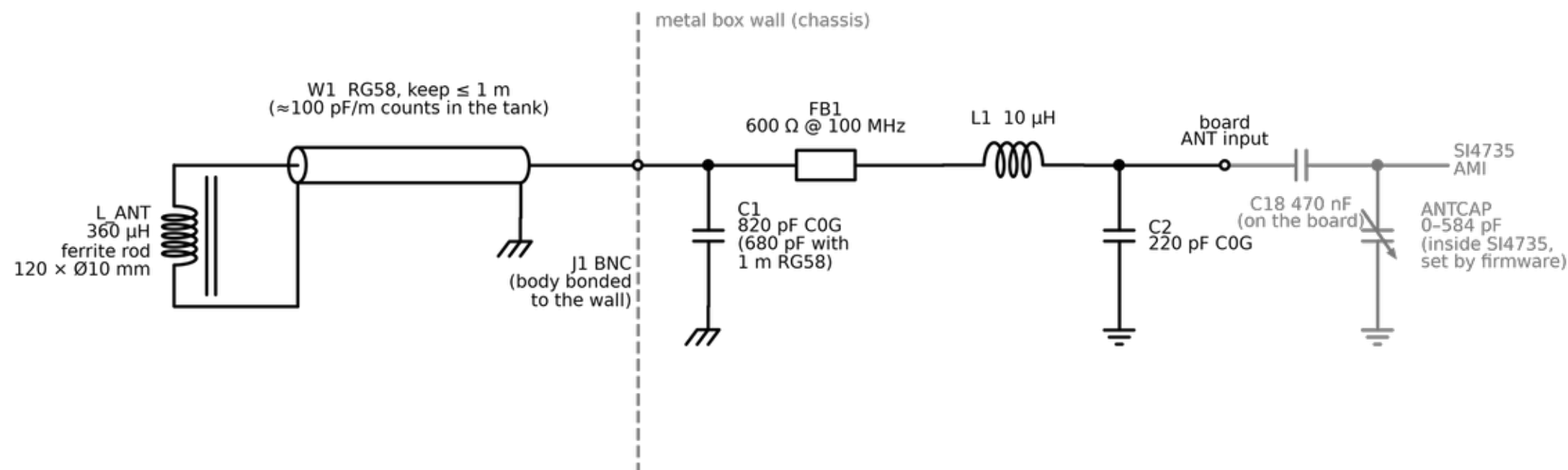


e-CzasPL receiver: 225 kHz ferrite antenna tank and VHF/UHF filter for the SI4735 AMI input

For firmware 2.0.4+ (automatic ANTCAP trim at power-up). All capacitors in the tank are C0G/NP0.



Resonance budget: 360 µH needs 1390 pF total at 225 kHz. ±10 % on L_ANT needs 1.26-1.54 nF, still inside the ANTCAP range in both variants.

	fixed C (incl. ~30 pF stray)	ANTCAP for 225 kHz	tuning range with ANTCAP 0-584 pF
A: antenna on the box, short leads (C1 = 820 pF)	1070 pF	≈ 320 pF	206-256 kHz
B: ~1 m RG58 to the antenna (C1 = 680 pF)	1030 pF	≈ 360 pF	209-261 kHz

1. Tank capacitors C1, C2 must be C0G/NP0: X7R changes by up to ±15 % with temperature and would detune the antenna by more than its 3-6 kHz bandwidth.
2. C1 goes directly at J1, to the wall, with the shortest possible leads: it carries the VHF/UHF current to the chassis. Bond the coax shield / cold end 360° at the wall.
3. FB1 and L1 are practically a short at 225 kHz (L1 = 14 Ω); with C1/C2 they form a π low-pass: roughly 40-60 dB at FM/DAB/TV, limited by layout.
4. C2 sits next to the board's ANT input; its ground is the board ground, joined to the chassis at one point near J1. C18 (470 nF, on the board) stays.
5. At power-up, firmware 2.0.4 sweeps ANTCAP and logs e.g. 'SI4735: antenna tuned, ANTCAP 345.2 pF'. If it logs 'no antenna resonance', read the RSSI list:
rising towards the right end = too little fixed C, add 100-150 pF to C1; rising towards the left end = too much, remove 100-150 pF.
6. Orientation: rod axis at right angles to the direction of Solec Kujawski. Keep the rod away from switching supplies and digital wiring.
7. Protection diodes, if ever needed, go after L1 (never before the low-pass), so they cannot rectify the strong VHF signals.

Parts list: TME (tme.eu) symbols

Ref	Qty	Part	TME symbol (manufacturer part)	Alternatives at TME
C1	1	820 pF C0G 50 V ± 5 %, 1206	C1206C821J5GACTU (KEMET)	GRM1885C1H821JA01D (Murata, 0603)
C1 (var. B)	1	680 pF C0G 50 V ± 5 %	CL10C681JB8NNNC (Samsung, 0603)	RCE5C1H681J0A2H03B (Murata, THT 2.5 mm)
C2	1	220 pF C0G 50 V ± 5 %, 1206	C1206C221J5GAC (KEMET)	CC1206JRNPO9221 (Yageo), CL31C221JBCNNNC (Samsung)
FB1	1	Ferrite bead 600 Ω @ 100 MHz, 1206	BLM31PG601SN1L (Murata)	LCBC-601 (Ferrocore)
L1	1	Inductor 10 μ H, 1210	LQH32CN100K53L (Murata)	LQH32PN100MNCL (Murata)
J1	1	BNC jack, bulkhead front mount, solder cup, non-isolated (body to the wall)	112575 (Amphenol RF)	B6351B1-ND3G-50 (Amphenol RF)
P1	1	BNC plug for RG58, crimp	BNC-203	R141-082-000 (Radiall)
W1	≤ 1 m	Coax RG58 50 Ω	TAS-RG58CU (Tasker RG58C/U)	SIVA-RG58CU (Siva Cavi)
L_ANT	1	Ferrite rod antenna 360 μ H, 120 $\times$ $\varnothing$ 10 mm	(existing)	

Stock was not verified: TME's site does not allow automated stock checks, and some 1206 C0G values have large minimum order quantities. Check the symbols in the TME basket; any C0G/NP0 50 V part of the same value and size is equivalent.

Mounting: C1 at J1 on the inside of the wall; FB1, L1, C2 on a small piece of copper-clad board between J1 and the receiver's ANT input, with the copper as ground bonded to the wall at J1.